



PRODUCT
DATASHEET '25-26'

ADVANCE ANALYTIK

REVOLUTIONIZING ONLINE MONITORING SOLUTIONS



OPTICS 1000 – PHOSPHATE

PHOSPHATE LOW RANGE (PO_4^{3-})

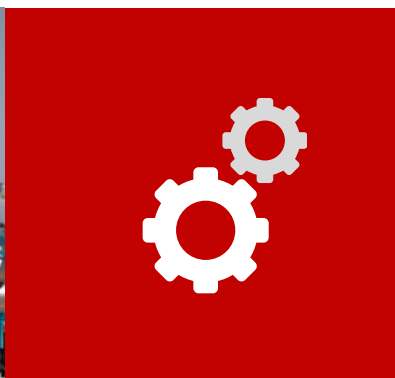
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Viz Series

PHOSPHATE LR (PO_4^{3-})

Method - Colorimetric

After adding the sample into the measurement cell, some reagents are added in order to adjust the solution to the desired conditions (pH, valence's elements, etc.). Then, a blank is done to correct any temperature or turbidity disturbance. Subsequently, a last reagent is added, and it reacts with solution developing a color, which is measured using a correct wavelength. Thanks to the photometer used, the result achieves a great accuracy



Principle of measurement

Ammonium molybdate and potassium antimonyl tartrate react in acid medium with orthophosphate to form a heteropoly acid-phosphomolybdic acid-that is reduced to intensely colored molybdenum blue by ascorbic acid.

Advantages of the method

The blue color developed is so intense and it provides a high accuracy and repeatability. The method allows to measure really low values of concentrations (lower than 50 ppb).



Interferences

Arsenates concentrations as low as 0.1 mg As/L interfere with the phosphate determination. Cr (VI) and NO₂⁻ interfere to give results about 3% low at concentrations of 1 mg/L and 10 to 15% low at 10 mg/L. Sulfide (Na₂S) and silicate do not interfere at concentrations of 1.0 and 10 mg/L.

Specifications

RANGE	From 0 to 200 ppb / 500 ppb / 1000 ppb. Higher concentrations using high range method.
ACCURACY	±2%
FULL SCALE REPEATABILITY	±2%
RESOLUTION	0,1 ppb
LOW LIMIT DETECTION	1ppb
ANALYSIS TIME	around 12 minutes
CALIBRATION	two point
LED WAVELENGTH	810 nm

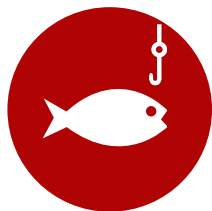
Reagents consumption

- Reagent 1: 0.5 ml / analysis - 0.5L / month
- Reagent 2: 0.5 ml / analysis - 0.5L / month

Monthly consumption calculated assuming 1 analysis per hour



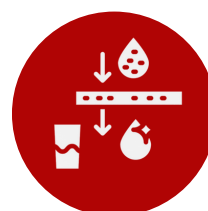
Applications



Fish farming



Food and Beverage



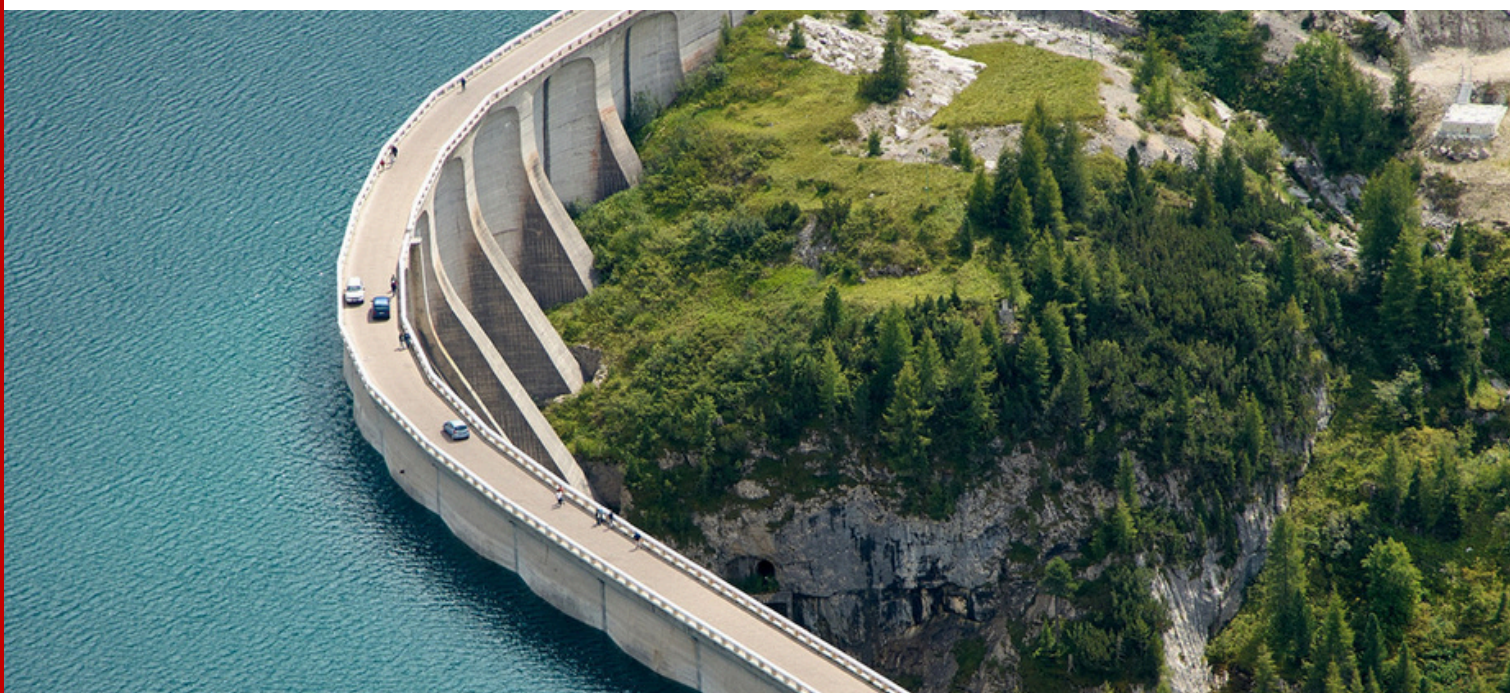
Desalination Plants



**Chemical Processing
Plants**



**Textile and Dyeing
Industries**



Note -

This data sheet serves as general information about the Optics 1000 - Phosphate (LR) . For specific technical details, installation guidelines, and troubleshooting assistance, please refer to the official user manual provided with the product.

For inquiries and detailed technical information, please contact sales@advanceanalytik.com.



LET'S CONNECT !

Let's work together to find a solution that works for you



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